TOPOLOGICAL INVARIANCE OF THE ETALE SITE RESTRICTS TO THE SMALL ETALE SITES OF SCHEMES AND OF AFFINE SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1414-1422 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 1424-1433, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The topological invariance theorem of the same chapter (theorem-topological-invariance, already collected as S3-spaces-topological-invariance) supplying the equivalence, and descent results for being a scheme and for affineness along integral universally bijective morphisms (Limits of Spaces); cited inputs with the argument written in the retained span.

Source reference. Original lemma 07VV: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1433–1433. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07VT: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1433–1433. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05ZH: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1316–1326. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *With assumption and notation as in Theorem 05ZH the equivalence of categories $Y_{spaces,étale} \to X_{spaces,étale}$ restricts to equivalences of categories $Y_{étale} \to X_{étale}$ and $Y_{affine,étale} \to X_{affine,étale}$.*

Proof. This is just the statement that given an object $V \in Y_{spaces,étale}$ we have V is a(n affine) scheme if and only if $V \times_Y X$ is a(n affine) scheme. Since $V \times_Y X \to V$ is integral, universally injective, and surjective (as a base change of $X \to Y$) this follows from Limits of Spaces, Lemma 07VV and Proposition 07VT. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.